

PPT Map Composition Is Entanglement Breaking When One Dimension Is Two

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The positive partial transpose (PPT) squared conjecture asks whether applying a completely positive and completely copositive map twice must produce an entanglement-breaking map. We prove its equivalent two-map formulation whenever the input, intermediate, or output matrix algebra is M_2 , with the other dimensions arbitrary and independent. In the state formulation, we also prove that every postselected measurement outcome on the intermediate systems of two PPT states is separable across the end systems if any one of the four local systems is a qubit. We also show that the composition of any two composable PPT maps is 2-entanglement breaking, as is its adjoint. Schmidt-number iteration then gives a uniform bound: any $2(d-1)$ PPT maps on M_d compose to an entanglement-breaking map.

Introduction.—Quantum channels are composed whenever a noisy process is used repeatedly or several elementary links are connected in a network. Entanglement-breaking maps mark the point at which a channel can no longer transmit entanglement: acting on one part of any bipartite state always produces a separable operator [1]. Positive partial transpose (PPT) maps form a larger class. Their Choi operators have positive partial transpose but may still be entangled, just as PPT states may contain bound entanglement [2]. Determining when PPT maps become entanglement breaking under composition therefore probes how repeated noise erases quantum correlations and whether one-shot entanglement swapping can entangle the end systems of two PPT links in a repeater network [3, 4].

The PPT squared conjecture asks what happens after two successive maps [5]. Its original form asks whether $T \circ T$ is entanglement breaking for every PPT map $T : M_d \rightarrow M_d$, without assuming trace preservation. Christandl, Müller-Hermes, and Wolf proved that this is equivalent to the corresponding statement for two possibly different composable PPT maps $T_1 : M_{d_0} \rightarrow M_{d_1}$ and $T_2 : M_{d_1} \rightarrow M_{d_2}$. The Choi–Jamiolkowski correspondence relates this formulation to entanglement swapping. Placing the two Choi operators on adjacent links and projecting their intermediate systems onto the standard maximally entangled state produces the Choi operator of $T_2 \circ T_1$ [6, 7]. This identifies map composition with the standard maximally entangled postselection in a single entanglement-swapping step and raises the broader question whether an arbitrary postselected measurement can entangle the end systems of two PPT links.

Two lines of work describe the progress toward this question. For repeated applications of one channel, entanglement-breaking indices quantify the number of uses required to destroy all entanglement [8]. Kennedy, Manor, and Paulsen proved asymptotic convergence to the cone of entanglement-breaking maps for unital or trace-preserving PPT maps [9]. Finite indices were then established for unital PPT maps and for PPT maps whose Perron eigenmatrices have full support [10, 11]. Park recently proved that every PPT map has finite entanglement-breaking index and asked whether the index admits a bound depending only on the dimension [12].

For the two-map formulation on M_d , the conjecture is immediate for $d = 2$ and was proved for $d = 3$ by Christandl,

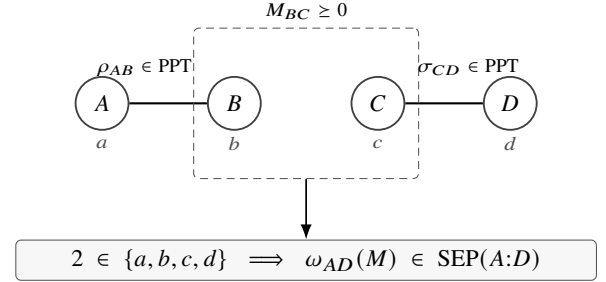


FIG. 1. One-shot entanglement swapping between two PPT states. For arbitrary local dimensions and any positive measurement operator M_{BC} on the intermediate systems, the unnormalized operator on $A:D$ is separable if $2 \in \{a, b, c, d\}$. The operator M_{BC} need not be rank one or projective, and b and c need not be equal.

Müller-Hermes, and Wolf and independently by Chen, Yang, and Tang [5, 13, 14]. Further results cover Gaussian channels, Choi-type maps, and stronger qutrit composition statements [5–7]. Henry very recently proved the case $M_{d_0} \rightarrow M_2 \rightarrow M_{d_2}$ for arbitrary d_0 and d_2 [15].

In this work, we prove that $T_2 \circ T_1$ is entanglement breaking whenever any one of d_0, d_1, d_2 equals two, with no relation required between the other dimensions. We establish $d_2 = 2$ using an auxiliary operator with a qutrit factor and a normal form for positive maps with two-dimensional domain, and obtain $d_0 = 2$ by Hilbert–Schmidt adjunction. A separate argument in the Pauli representation gives an independent proof of the case $d_1 = 2$. We further prove that every postselected outcome of two PPT links is separable across the end systems whenever any one of the four local systems is a qubit, even when the measured dimensions differ.

The proof for $d_2 = 2$ also implies that both the composition of any two PPT maps and its adjoint are 2-entanglement breaking. Schmidt-number iteration then shows that any $2(d-1)$ PPT maps on M_d , not necessarily identical, compose to an entanglement-breaking map. In particular, $n_{EB}(T) \leq 2(d-1)$ for every PPT map $T : M_d \rightarrow M_d$, answering Question 4.3 of Ref. [12].

Main result.— For a map Φ , use the Choi operator

$$J_\Phi = \sum_{i,j} E_{ij} \otimes \Phi(E_{ij}), \quad (1)$$

with partial transpose on the second factor. A map is PPT when its Choi operator is positive and PPT, and entanglement breaking (EB) when its Choi operator is separable [1, 16]. No map below is assumed trace preserving, unital, invertible, or of fixed rank.

Theorem 1. *Let*

$$\Phi_1 : M_{d_0} \longrightarrow M_{d_1}, \quad \Phi_2 : M_{d_1} \longrightarrow M_{d_2} \quad (2)$$

be PPT maps. If $2 \in \{d_0, d_1, d_2\}$, then $\Phi_2 \circ \Phi_1$ is entanglement breaking. The other two dimensions are arbitrary and independent.

The proof has two branches. We establish $d_2 = 2$ directly and obtain $d_0 = 2$ by adjunction; an independent Pauli argument proves $d_1 = 2$.

Composition with a two-dimensional output algebra.— When the output algebra is M_2 , the adjacent $2 \otimes d$ PPT Choi operator may itself be entangled. Separability of the composed Choi operator is instead equivalent to the following bilinear positivity statement. Here $\text{PPT}(n, d)$ denotes the cone of positive operators on $\mathbb{C}^n \otimes \mathbb{C}^d$ whose partial transpose on the second factor is positive.

Lemma 2. *For all $n, d \geq 1$, every positive map $T : M_2 \rightarrow M_d$, every $X \in \text{PPT}(n, d)$, and every $R \in \text{PPT}(n, 2)$,*

$$\text{Tr}[X(\text{id}_n \otimes T)(R)] \geq 0. \quad (3)$$

The quantifier over all positive T is essential: their Choi operators form the cone of block-positive operators. The proof constructs an operator Q with a qutrit factor from R and then reduces every such T to positive maps whose restrictions to the relevant two-dimensional subspaces are decomposable.

Lemma 3. *For every $R \in \text{PPT}(n, 2)$, there exists a positive operator $Q \in M_n \otimes M_3$ satisfying $Q^{\Gamma_{\text{qutrit}}} = Q$, where the superscript transposes the qutrit factor. It obeys*

$$(\text{id}_n \otimes \text{Ad}_V)(Q) = R, \quad V = \begin{pmatrix} 1 & -i & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad (4)$$

where $\text{Ad}_L(Z) = LZL^*$. The operator Q also has a pure-state decomposition in which every vector has Schmidt rank at most two and its Schmidt vectors on $\mathbb{C}^3$ lie in a conjugation-invariant subspace of dimension at most two.

The operator Q can be constructed explicitly. Writing R as a 2×2 matrix of blocks in M_n , $R = \begin{pmatrix} A & C \\ C^* & B \end{pmatrix}$, with $C = X + iY$, $X = X^*$, and $Y = Y^*$, determines

$$\begin{aligned} G &= B^+, & \Delta &= A - XGX - YGY, \\ W &= YGX - XGY, & \Xi &= XGY + YGX. \end{aligned} \quad (5)$$

For $P_B = \text{supp } B$, generalized Schur complements give $X = P_B X P_B$, $Y = P_B Y P_B$, and $\Delta \pm iW \geq 0$. Define $\tilde{Q} \in M_3 \otimes M_n$ by

$$\tilde{Q} = \begin{pmatrix} XGX + \Delta/2 & -\Xi/2 & X \\ -\Xi/2 & YGY + \Delta/2 & -Y \\ X & -Y & B \end{pmatrix}, \quad Q = F_{3,n} \tilde{Q} F_{3,n}^*, \quad (6)$$

where $F_{3,n}$ flips the tensor factors. Direct multiplication gives Eq. (4), and symmetry of the 3×3 block matrix gives $Q^{\Gamma_{\text{qutrit}}} = Q$. The operator $\tilde{Q}$ has the positive splitting

$$\begin{aligned} \tilde{Q} &= Q_0 + Q_{12}, \\ Q_0 &= \begin{pmatrix} X \\ -Y \\ B \end{pmatrix} G \begin{pmatrix} X & -Y & B \end{pmatrix}, \\ Q_{12} &= \frac{1}{2} \begin{pmatrix} \Delta & -W & 0 \\ W & \Delta & 0 \\ 0 & 0 & 0 \end{pmatrix}. \end{aligned} \quad (7)$$

For $v_\pm = (e_0 \pm ie_1)/\sqrt{2}$,

$$Q_{12} = \frac{1}{2} [|v_+\rangle\langle v_+| \otimes (\Delta - iW) + |v_-\rangle\langle v_-| \otimes (\Delta + iW)] \geq 0, \quad (8)$$

and its pure components lie in the subspace $\text{span}\{e_0, e_1\}$. The first term in Eq. (7) is a Gram operator. On $\text{supp } B$, put $S = B^{1/2}$, $D = B^{+1/2} X B^{+1/2}$, and $F = B^{+1/2} Y B^{+1/2}$. Diagonalizing the Hermitian matrix $De_j = d_j e_j$ gives, after the tensor flip, Gram vectors

$$|\xi_j\rangle = S e_j \otimes (d_j e_0 + e_2) - S F e_j \otimes e_1, \quad d_j \in \mathbb{R}. \quad (9)$$

Their Schmidt vectors on $\mathbb{C}^3$ lie in the conjugation-invariant subspace $\text{span}\{d_j e_0 + e_2, e_1\}$. Thus both terms in Eq. (7) are positive and have the decomposition stated in Lemma 3. The treatment of singular B is given in Ref. [17].

This restriction implies the required nonnegative pairing. If a positive map $\Lambda : M_3 \rightarrow M_k$ satisfies $\Lambda = \Lambda \circ \text{Sym}$, with $\text{Sym}(Z) = (Z + Z^T)/2$, then every real isometry $U : \mathbb{C}^2 \rightarrow \mathbb{C}^3$ obeys $\Lambda \circ \text{Ad}_U = (\Lambda \circ \text{Ad}_U) \circ T_2$. The restriction is therefore decomposable, by separability of positive $2 \otimes k$ operators fixed by qubit partial transpose [18, Theorem 2] and duality between the PPT cone and decomposable witnesses [19]. To apply this fact termwise, write the decomposition from Lemma 3 as $Q = \sum_\alpha |\psi_\alpha\rangle\langle\psi_\alpha|$. For each component, choose a real isometry $U_\alpha : \mathbb{C}^2 \rightarrow \mathbb{C}^3$ whose range contains its Schmidt vectors on $\mathbb{C}^3$ and a map $C_\alpha : \mathbb{C}^n \rightarrow \mathbb{C}^2$ such that

$$\begin{aligned} |\psi_\alpha\rangle &= (I_n \otimes U_\alpha) \text{vec}(C_\alpha), \\ (\text{id}_n \otimes \Lambda)(|\psi_\alpha\rangle\langle\psi_\alpha|) &= J_{(\Lambda \circ \text{Ad}_{U_\alpha}) \circ \text{Ad}_{C_\alpha}}. \end{aligned} \quad (10)$$

Decomposable maps remain decomposable under CP precomposition; for the copositive part this follows from $T_2 \circ \text{Ad}_{C_\alpha} = \text{Ad}_{C_\alpha} \circ T_n$. Hence every term in Eq. (10) pairs nonnegatively with a PPT operator, and

$$\text{Tr}[P(\text{id}_n \otimes \Lambda)(Q)] \geq 0 \quad \text{for every } P \in \text{PPT}(n, k). \quad (11)$$

We now reduce an arbitrary positive $T : M_2 \rightarrow M_d$ to the map Λ and CP maps to which Eq. (11) applies. Denote the left-hand side of Eq. (3) by $\mathcal{B}_T(X, R)$. The zero map is trivial. For $T \neq 0$, first restrict its codomain to the support of $T(I_2)$, which contains the range of every output. The pairing and the PPT constraints are invariant under invertible input and output congruences. Moving a pure input with nonzero output to e_0 , whitening that output on its range, and applying an upper shear gives the following block form. If the chosen output has rank k and $m = d - k$, then

$$\begin{aligned} T(E_{00}) &= \begin{pmatrix} I_k & 0 \\ 0 & 0_m \end{pmatrix}, & T(E_{01}) &= \begin{pmatrix} H & A \\ B^* & 0 \end{pmatrix}, \\ T(E_{11}) &= \begin{pmatrix} C & 0 \\ 0 & I_m \end{pmatrix}. \end{aligned} \quad (12)$$

When $m = 0$, the lower blocks are absent. The restriction to $\text{supp } T(I_2)$, the exclusion of common kernel vectors, and the treatment of singular blocks are given in Ref. [17]. From here on, A, B, C denote the blocks in Eq. (12), not the blocks defining Q above. For a real vector $x = (u, v, r)^T$, set

$$\begin{aligned} M(u, v) &= u(A + B) + iv(B - A), \\ \mathcal{S}(x) &= (u^2 + v^2)I_k - M(u, v)M(u, v)^* \\ &\quad + r[u(H + H^*) + iv(H^* - H)] + r^2C. \end{aligned} \quad (13)$$

Since $Vx = (u - iv, r)^T$, Eq. (12) gives

$$T((Vx)(Vx)^*) = \begin{pmatrix} \mathcal{S}(x) + M(u, v)M(u, v)^* & rM(u, v) \\ rM(u, v)^* & r^2I_m \end{pmatrix} \geq 0. \quad (14)$$

For $m > 0$ and $r \neq 0$, the lower-right Schur complement gives $\mathcal{S}(x) \geq 0$; continuity covers $r = 0$. For $m = 0$, the rectangular blocks and $M(u, v)$ are absent and the same conclusion follows directly.

Real polarization of $\mathcal{S}$, followed by complexification and composition with Sym , defines a complex-linear map $\Lambda : M_3 \rightarrow M_k$ satisfying

$$\Lambda(xx^T) = \mathcal{S}(x), \quad \Lambda = \Lambda \circ \text{Sym}. \quad (15)$$

This map is positive: if $Z \geq 0$, then $\text{Sym}(Z) = (Z + \bar{Z})/2$ is a real positive matrix and hence a sum $\sum_\ell x_\ell x_\ell^T$ with $x_\ell \in \mathbb{R}^3$; Eq. (15) gives $\Lambda(Z) = \sum_\ell \mathcal{S}(x_\ell) \geq 0$.

Let $E : \mathbb{C}^k \hookrightarrow \mathbb{C}^d$ be the canonical inclusion into the first k coordinates. For the standard basis e_a of $\mathbb{C}^m$ and $a = 1, \dots, m$, define

$$K_a = \begin{pmatrix} (A+B)e_a & i(B-A)e_a & 0 \\ 0 & 0 & e_a \end{pmatrix} : \mathbb{C}^3 \rightarrow \mathbb{C}^d. \quad (16)$$

Then, abbreviating $M = M(u, v)$,

$$\begin{aligned} K_a x &= \begin{pmatrix} M e_a \\ r e_a \end{pmatrix}, \\ \sum_{a=1}^m (K_a x)(K_a x)^* &= \begin{pmatrix} M M^* & r M \\ r M^* & r^2 I_m \end{pmatrix}. \end{aligned} \quad (17)$$

For $m = 0$ the family is absent. Adding $\text{Ad}_E(\Lambda(xx^T))$ to Eq. (17) reproduces Eq. (14) for every real x ; polarization therefore gives the exact map identity

$$T \circ \text{Ad}_V \circ \text{Sym} = \text{Ad}_E \circ \Lambda + \sum_{a=1}^m \text{Ad}_{K_a} \circ \text{Sym}. \quad (18)$$

The treatment of the support of $T(I_2)$ is given in Ref. [17]. Since $Q^{\Gamma_{\text{qutrit}}} = Q$, inserting Eq. (18) into Eq. (3) yields the identity

$$\begin{aligned} \text{Tr}[X(\text{id}_n \otimes T)(R)] &= \text{Tr}[(\text{id}_n \otimes \text{Ad}_{E^*})(X)(\text{id}_n \otimes \Lambda)(Q)] \\ &\quad + \sum_a \text{Tr}[(\text{id}_n \otimes \text{Ad}_{K_a^*})(X)Q] \geq 0. \end{aligned} \quad (19)$$

The first compressed operator is PPT, while every operator in the sum is positive; Eq. (11) proves Lemma 2.

The bilinear inequality proves the case of a two-dimensional output algebra. For PPT maps $\Phi : M_d \rightarrow M_n$ and $\Psi : M_n \rightarrow M_2$, adjunction preserves PPT, and therefore $X = J_{\Phi^*} \in \text{PPT}(n, d)$ and $R = J_\Psi \in \text{PPT}(n, 2)$. A matrix-unit contraction gives

$$\begin{aligned} \mathcal{B}_T(J_{\Phi^*}, J_\Psi) &= \text{Tr}_{\text{HS}}(T \circ \Psi \circ \Phi) \\ &= \text{Tr}[J_{T^*} J_{\Psi \circ \Phi}], \end{aligned} \quad (20)$$

where $\text{Tr}_{\text{HS}} S = \sum_{i,j} \text{Tr}[E_{ji} S(E_{ij})]$. As $T : M_2 \rightarrow M_d$ ranges over all positive maps, J_{T^*} exhausts the cone of block-positive operators. Lemma 2 makes Eq. (20) nonnegative for every such operator. Duality between the cone of block-positive operators and the cone of separable operators [13] then makes $J_{\Psi \circ \Phi}$ separable. Hence $\Psi \circ \Phi$ is EB. Applying the same argument to the adjoint PPT pair proves the case of a two-dimensional input algebra.

Two-dimensional intermediate algebra.— Very recently, Henry proved the case with a two-dimensional intermediate algebra using normalized 2×2 Choi blocks and operator interpolation [15]. We give an independent proof using qubit spin flip and the following separability identity.

Lemma 4. *For Hermitian $X_j \in M_a$ and $Y_j \in M_c$, if*

$$\sum_j X_j^2 \leq I_a, \quad \sum_j Y_j^2 \leq I_c, \quad (21)$$

then $I_a \otimes I_c + \sum_j X_j \otimes Y_j$ is separable.

The explicit decomposition into positive product operators is given in Ref. [17].

For $d_1 = 2$, set $\Psi = \Phi_1^*$ and $\Theta = \Phi_2$, and whiten them on their output supports to unital PPT maps $\hat{\Psi}$ and $\hat{\Theta}$. Define $X_j = \hat{\Psi}(\sigma_j)$ and $Y_j = \hat{\Theta}(\sigma_j)$ for the three Pauli matrices. Complete positivity, complete copositivity, and the qubit spin flip give

$$-I \leq \sum_j \sigma_j^T \otimes X_j \leq I. \quad (22)$$

Functional calculus followed by the normalized qubit trace yields $\sum_j X_j^2 \leq I$; the same holds for the Y_j . Hence

$$J_{\widehat{\Theta} \circ \widehat{\Psi}^*} = \frac{1}{2} \left(I \otimes I + \sum_{j=1}^3 X_j^T \otimes Y_j \right) \quad (23)$$

is separable by Lemma 4. Local CP congruences undo the whitening and preserve separability, proving the case $d_1 = 2$. Singular support details are given in Ref. [17]. Together with the input and output cases above, this completes the proof of Theorem 1.

Entanglement swapping and consequences for composition.— The map theorem has a direct consequence for states. For the two PPT states in Fig. 1, a positive measurement operator $M_{BC} \geq 0$ on the intermediate systems produces the unnormalized operator on the end systems

$$\omega_{AD}(M) = \text{Tr}_{BC}[(\rho_{AB} \otimes \sigma_{CD})(I_A \otimes M_{BC} \otimes I_D)]. \quad (24)$$

Corollary 5. *Let $\dim A = a$, $\dim B = b$, $\dim C = c$, and $\dim D = d$, with no equality required among these dimensions. If $\rho_{AB} \in \text{PPT}(A : B)$, $\sigma_{CD} \in \text{PPT}(C : D)$, and $2 \in \{a, b, c, d\}$, then*

$$\omega_{AD}(M) \in \text{SEP}(A : D) \quad \text{for every } M_{BC} \geq 0. \quad (25)$$

For unequal intermediate dimensions, a rank-one vector can be represented in two ways using rectangular matrices,

$$|\eta\rangle = (L_B \otimes I_C)|\Omega_c\rangle = (I_B \otimes L_C)|\Omega_b\rangle, \quad (26)$$

where $L_B : C' \rightarrow B$, $L_C = L_B^T : B' \rightarrow C$, and the vectors $|\Omega_c\rangle$ and $|\Omega_b\rangle$ are unnormalized maximally entangled vectors on $C':C$ and $B:B'$, respectively. Moving the rectangular filters onto the two PPT states reduces either expression to a standard maximally entangled projection. The two choices cover a qubit at A, C, D and at B , respectively. Spectral decomposition of M then proves Corollary 5; the two identities used above, singular filters, and local CP processing are treated in Ref. [17].

A CP map $K : M_p \rightarrow M_q$ is *2-entanglement breaking* (2-EB) when $(\text{id}_2 \otimes K)(Z)$ is separable for every positive $Z \in M_2 \otimes M_p$ [5, Definition II.2]. Equivalently, $K \circ E$ is EB for every CP map $E : M_2 \rightarrow M_p$ [20, Theorem 3.5].

Corollary 6. *Let $\Phi_1 : M_{d_0} \rightarrow M_{d_1}$ and $\Phi_2 : M_{d_1} \rightarrow M_{d_2}$ be PPT. Then both $H = \Phi_2 \circ \Phi_1$ and $H^* = \Phi_1^* \circ \Phi_2^*$ are 2-EB.*

For any CP $E_0 : M_2 \rightarrow M_{d_0}$, the map $\Phi_1 \circ E_0$ is PPT, and the case $d_0 = 2$ makes $H \circ E_0$ EB. Applying the same argument to the PPT pair (Φ_2^*, Φ_1^*) proves the assertion for H^* ; this separate application is needed because the class of 2-EB maps is not generally closed under adjoints [20, Theorem 3.17].

Corollary 7. *For $d \geq 2$, every composition of $2(d-1)$ PPT maps $T_j : M_d \rightarrow M_d$ is EB. Hence, for $n_{\text{EB}}(T) = \min\{n \geq 1 : T^n \in \text{EB}\}$,*

$$\sup_{T \in \text{PPT}(M_d, M_d)} n_{\text{EB}}(T) \leq 2(d-1). \quad (27)$$

Grouping the maps into $d-1$ adjacent pairs makes each pair 2-EB by Corollary 6. The Schmidt-number iteration theorem then makes their composition EB [5, Theorem II.1]. The factors need not be identical. Park obtains sharper dimension-independent bounds for certain subclasses of PPT maps; in contrast, Eq. (27) applies to all PPT maps and answers the question of a bound depending only on the dimension in Ref. [12, Question 4.3].

Discussion.— Dimension two gives the same conclusion at the input, intermediate, and output algebras, although the proof uses two different arguments. Hilbert–Schmidt adjoints interchange the input and output cases, which follow from one bilinear inequality. The intermediate case instead uses the simultaneous 2×2 structure of both Choi operators. The endpoint proof uses more than a bound on Schmidt rank. The operator Q need not be separable, and Schmidt rank two alone would leave the $2 \otimes d$ PPT obstruction intact.

The operational statement concerns one copy of each of two independent links, one relay use, and a discarded relay output. Local CP maps acting separately on the four systems and classical coarse-graining preserve it, but retained relay memory and adaptive or multicopy processing are outside its scope.

The problem without a two-dimensional algebra, including equal dimensions $d \geq 4$, remains open. Our result guarantees that every composition of two PPT maps is 2-EB, whereas the PPT squared conjecture asks whether it is EB. Extending the proof to higher dimensions would require an analogue of the two-dimensional subspaces invariant under complex conjugation used here.

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Supplemental Material for *PPT Map Composition Is Entanglement Breaking When One Dimension Is Two*

This Supplemental Material fixes the Choi and partial-transpose conventions and proves the statement for every positive measurement operator when the two intermediate dimensions differ. Sections S2–S4 give the complete proof for a two-dimensional input or output algebra, including all singular cases; Sec. S5 records the independent Pauli proof for a two-dimensional intermediate algebra, and Sec. S6 completes the proof of the composition theorem and derives the consequences. All matrix algebras are finite dimensional.

S1. CONVENTIONS, CONE DUALITY, AND POSITIVE MEASUREMENT OPERATORS

For a linear map $\Phi : M_a \rightarrow M_b$, we use the unnormalized Choi operator

$$J_\Phi = \sum_{i,j=0}^{a-1} E_{ij} \otimes \Phi(E_{ij}) \in M_a \otimes M_b. \quad (\text{S1})$$

Thus Φ is completely positive (CP) precisely when $J_\Phi \geq 0$, and it is entanglement breaking (EB) precisely when J_Φ is separable. We write T_a for transposition in the fixed computational basis and $Z^{\Gamma_A} = (\mathsf{T}_a \otimes \text{id})(Z)$. Our main convention is partial transpose on the second factor,

$$Z^{\Gamma_B} = (\text{id}_a \otimes \mathsf{T}_b)(Z). \quad (\text{S2})$$

The cone

$$\text{PPT}(a, b) = \{Z \geq 0 : Z^{\Gamma_B} \geq 0\} \quad (\text{S3})$$

matches the convention in the main text, where partial transpose acts on the output factor. Nothing depends on this choice: $Z^{\Gamma_A} = (Z^{\Gamma_B})^\top$, so positivity of either partial transpose is equivalent. A CP map is called PPT when its Choi operator belongs to this cone, with partial transpose on the output factor; equivalently, it is both completely positive and completely copositive. We use $\text{Ad}_L(X) = LXL^*$. The Hilbert–Schmidt adjoint is denoted by Φ^* .

A Hermitian operator $W \in M_a \otimes M_b$ is block positive when $\langle x \otimes y, W(x \otimes y) \rangle \geq 0$ for all x, y . Under the trace pairing, the cone of block-positive operators is dual to the cone of separable operators, whereas the cone of decomposable witnesses is dual to the PPT cone [13]. In particular, a map is positive exactly when its Choi operator is block positive.

Let $A \simeq \mathbb{C}^a$, $B \simeq \mathbb{C}^b$, $C \simeq \mathbb{C}^c$, and $D \simeq \mathbb{C}^d$, with no equality assumed among the four dimensions. For positive operators ρ_{AB} , σ_{CD} , and a positive measurement operator $M_{BC} \geq 0$ on the intermediate systems, the unnormalized operator on the end systems is

$$\begin{aligned} \omega_M(\rho, \sigma) &= \text{Tr}_{BC} \left[(I_A \otimes M_{BC}^{1/2} \otimes I_D)(\rho_{AB} \otimes \sigma_{CD})(I_A \otimes M_{BC}^{1/2} \otimes I_D) \right] \\ &= \text{Tr}_{BC} [(\rho_{AB} \otimes \sigma_{CD})(I_A \otimes M_{BC} \otimes I_D)]. \end{aligned} \quad (\text{S4})$$

The second equality is cyclicity on the traced systems. Physical POVM effects additionally satisfy $M_{BC} \leq I_{BC}$, but the distinction is immaterial here: every nonzero positive operator can be rescaled to an effect and $\omega_{\lambda M} = \lambda \omega_M$ for $\lambda > 0$, while $M = 0$ is trivial. Introduce copies $C' \simeq C$ and $B' \simeq B$, together with

$$\begin{aligned} |\Omega_c\rangle_{C'C} &= \sum_{q=0}^{c-1} |q\rangle_{C'} |q\rangle_C, & \Omega_c &= |\Omega_c\rangle\langle\Omega_c|, \\ |\Omega_b\rangle_{BB'} &= \sum_{p=0}^{b-1} |p\rangle_B |p\rangle_{B'}, & \Omega_b &= |\Omega_b\rangle\langle\Omega_b|. \end{aligned} \quad (\text{S5})$$

For equal intermediate dimensions, the passage from maximally entangled postselection to an arbitrary effect is the local-filtering argument of Singh and Nechita and of An and Lee [6, Proposition 2.1] [7, Theorem 9 and Appendix E]. For unequal dimensions, an arbitrary vector on the intermediate systems has the form

$$|\eta\rangle = \sum_{p=0}^{b-1} \sum_{q=0}^{c-1} \eta_{pq} |p\rangle_B |q\rangle_C \quad (\text{S6})$$

Define, in the chosen bases,

$$L = \sum_{p,q} \eta_{pq} |p\rangle_B \langle q|_{C'} : C' \rightarrow B, \quad R = L^\top : B' \rightarrow C. \quad (\text{S7})$$

Then

$$|\eta\rangle = (L \otimes \mathbf{I}_C) |\Omega_c\rangle = (\mathbf{I}_B \otimes R) |\Omega_b\rangle. \quad (\text{S8})$$

Cyclicity in Eq. (S4) gives

$$\begin{aligned} \omega_{|\eta\rangle\langle\eta|}(\rho, \sigma) &= \omega_{\Omega_c}((\mathbf{I}_A \otimes L^*)\rho(\mathbf{I}_A \otimes L), \sigma) \\ &= \omega_{\Omega_b}(\rho, (R^* \otimes \mathbf{I}_D)\sigma(R \otimes \mathbf{I}_D)). \end{aligned} \quad (\text{S9})$$

The first line is a maximally entangled contraction on $C':C$ and therefore corresponds to a composable pair with dimensions (a, c, d) . The second contracts $B:B'$ and corresponds to dimensions (a, b, d) .

Both rectangular filters preserve the PPT property, including when they are singular, because for local matrices of the indicated sizes

$$[(L_A \otimes L_B)Z(L_A \otimes L_B)^*]^{\Gamma_B} = (L_A \otimes \overline{L_B})Z^{\Gamma_B}(L_A \otimes \overline{L_B})^* \geq 0. \quad (\text{S10})$$

Moreover, if $\rho = J_{\Phi_1}$ and $\sigma = J_{\Phi_2}$, with $\Phi_1 : M_{d_0} \rightarrow M_{d_1}$ and $\Phi_2 : M_{d_1} \rightarrow M_{d_2}$, then a matrix-unit contraction gives

$$\omega_{\Omega_{d_1}}(J_{\Phi_1}, J_{\Phi_2}) = J_{\Phi_2 \circ \Phi_1}. \quad (\text{S11})$$

When $b = c = d_1$, Eq. (S9) reduces to Eq. (S11).

Now assume $\rho_{AB} \in \text{PPT}(a, b)$ and $\sigma_{CD} \in \text{PPT}(c, d)$. If the qubit is at A , C , or D , the first line of Eq. (S9) and Theorem 1 make every rank-one outcome separable. If the qubit is at B , use the second line. Thus every rank-one measurement outcome gives a separable operator whenever $2 \in \{a, b, c, d\}$. Finally diagonalize $M = \sum_\ell \lambda_\ell |\eta_\ell\rangle\langle\eta_\ell|$, with $\lambda_\ell \geq 0$. Linearity of ω_M and convexity of the cone of separable operators prove the statement for every positive effect without any equality between b and c .

The same conclusion is stable under local trace-nonincreasing CP processing around this single relay use. Let

$$\mathcal{A} : M_A \rightarrow M_{\tilde{A}}, \quad \mathcal{B} : M_B \rightarrow M_{\tilde{B}}, \quad \mathcal{C} : M_C \rightarrow M_{\tilde{C}}, \quad \mathcal{D} : M_D \rightarrow M_{\tilde{D}}, \quad (\text{S12})$$

be local trace-nonincreasing CP maps applied before the relay operation. A selected branch of the relay instrument is a trace-nonincreasing CP map $\mathcal{R}_r : M_{\tilde{B}} \otimes M_{\tilde{C}} \rightarrow M_R$. Once its quantum output R is discarded, its contribution to the operator on the end systems depends only on the positive operators

$$E_r = \mathcal{R}_r^*(\mathbf{I}_R) \geq 0, \quad M_r = (\mathcal{B}^* \otimes \mathcal{C}^*)(E_r) \geq 0. \quad (\text{S13})$$

If $\mathcal{A}'_r$ and $\mathcal{D}'_r$ are outcome-conditioned local trace-nonincreasing CP maps on the processed end systems, the unnormalized operator for outcome r is

$$\tau^{(r)} = [(\mathcal{A}'_r \circ \mathcal{A}) \otimes (\mathcal{D}'_r \circ \mathcal{D})][\omega_{M_r}(\rho, \sigma)]. \quad (\text{S14})$$

Equation (S14) follows from the defining trace pairing for the adjoints and shows that all processing at the relay is absorbed into the positive operator M_r on the original systems $B \otimes C$. For PPT ρ and σ , Corollary 5 makes $\omega_{M_r}(\rho, \sigma)$ separable whenever $2 \in \{a, b, c, d\}$. Local CP maps preserve separability, and hence $\tau^{(r)} \in \text{SEP}$ for every branch. Every nonzero branch remains separable after normalization. Classical coarse-graining also preserves the conclusion because

$$\tau^{(S)} = \sum_{r \in S} \tau^{(r)} \in \text{SEP}, \quad (\text{S15})$$

for any set S of relay outcomes.

This statement concerns one copy of each of two initially independent PPT links, one relay use, and the operator on the end systems after the relay quantum output has been discarded. It permits dimension-changing local maps but does not include quantum communication from the relay, retained relay quantum memory, later adaptive relay uses, or multicopy processing. Here “local” means that each map acts only on its labeled subsystem; joint operations across $A:B$, $C:D$, or $A:D$ are outside the statement. The qubit condition refers to the original link dimensions a, b, c, d , not to the dimensions after local processing.

S2. CONSTRUCTION AND DECOMPOSITION OF THE AUXILIARY OPERATOR

This section constructs the auxiliary operator used when the input or output algebra is two dimensional. Let $R \in \text{PPT}(n, 2)$. We regard R as a 2×2 matrix of blocks in M_n , and later introduce $\tilde{Q}$ as a 3×3 matrix of blocks in M_n . Write

$$R = \begin{pmatrix} A & C \\ C^* & B \end{pmatrix}, \quad C = X + iY, \quad X = X^*, \quad Y = Y^*, \quad G = B^+. \quad (\text{S16})$$

For a positive block matrix with lower-right block B , the generalized Schur complement gives $C = CP_B$ and $A - CB^+C^* \geq 0$, where $P_B = \text{supp } B$. Applying this statement also to $R^{\Gamma_{\text{qubit}}} = \begin{pmatrix} A & C^* \\ C & B \end{pmatrix}$, where the qubit is the second factor of $R \in M_n \otimes M_2$, gives $C = P_B CP_B$. Hence

$$X = P_B X P_B, \quad Y = P_B Y P_B. \quad (\text{S17})$$

Set

$$N = XGX + YGY, \quad W = YGX - XGY, \quad \Delta = A - N. \quad (\text{S18})$$

Since

$$CB^+C^* = N + iW, \quad C^*B^+C = N - iW, \quad (\text{S19})$$

the two generalized Schur complements are exactly

$$\Delta - iW \geq 0, \quad \Delta + iW \geq 0. \quad (\text{S20})$$

The operator $\tilde{Q} \in M_3 \otimes M_n$ is

$$\tilde{Q} = \begin{pmatrix} XGX + \Delta/2 & -(XGY + YGX)/2 & X \\ -(XGY + YGX)/2 & YGY + \Delta/2 & -Y \\ X & -Y & B \end{pmatrix}. \quad (\text{S21})$$

Let $F_{3,n} : \mathbb{C}^3 \otimes \mathbb{C}^n \rightarrow \mathbb{C}^n \otimes \mathbb{C}^3$ be the tensor flip and set

$$Q = F_{3,n} \tilde{Q} F_{3,n}^* \in M_n \otimes M_3. \quad (\text{S22})$$

The 3×3 matrix of $n \times n$ blocks is symmetric under transposition of the M_3 factor. Hence $\tilde{Q}$ is fixed by partial transpose on its first factor. Equivalently, Q satisfies

$$Q^{\Gamma_{\text{qutrit}}} = Q. \quad (\text{S23})$$

For

$$V = \begin{pmatrix} 1 & -i & 0 \\ 0 & 0 & 1 \end{pmatrix} : \mathbb{C}^3 \rightarrow \mathbb{C}^2, \quad (\text{S24})$$

direct block multiplication gives

$$(\text{id}_n \otimes \text{Ad}_V)(Q) = R. \quad (\text{S25})$$

Indeed, in $\tilde{Q}$ the $(0, 0)$ output block is the sum of the first two diagonal blocks, the $(0, 1)$ block is $X + iY = C$, and the $(1, 1)$ block is B .

Lemma S1. *The operator Q in Eq. (S22) is positive and admits a decomposition*

$$Q = \sum_{\alpha} |\psi_{\alpha}\rangle \langle \psi_{\alpha}| \quad (\text{S26})$$

such that every ψ_{α} has Schmidt rank at most two and its Schmidt vectors on $\mathbb{C}^3$ are contained in a conjugation-invariant subspace of dimension at most two in $\mathbb{C}^3$.

Proof. It is equivalent to work with $\tilde{Q}$, which splits as $\tilde{Q} = Q_0 + Q_{12}$, where

$$Q_0 = \begin{pmatrix} X \\ -Y \\ B \end{pmatrix} G \begin{pmatrix} X & -Y & B \end{pmatrix}, \quad (\text{S27})$$

$$Q_{12} = \frac{1}{2} \begin{pmatrix} \Delta & -W & 0 \\ W & \Delta & 0 \\ 0 & 0 & 0 \end{pmatrix}. \quad (\text{S28})$$

Let $v_{\pm} = (e_0 \pm ie_1)/\sqrt{2}$. Equation (S20) gives

$$Q_{12} = \frac{1}{2} [|v_+\rangle\langle v_+| \otimes (\Delta - iW) + |v_-\rangle\langle v_-| \otimes (\Delta + iW)] \geq 0. \quad (\text{S29})$$

Factoring the two positive operators on the second tensor factor into rank-one terms produces pure vectors whose qutrit components lie in the subspace $\text{span}_{\mathbb{C}}\{e_0, e_1\}$.

If $B = 0$, Eq. (S17) gives $X = Y = 0$, hence $Q_0 = 0$, and Eq. (S29) already proves the lemma. It remains to treat Q_0 on the nonzero support $P_B\mathbb{C}^n$.

For Q_0 , put

$$L = B^{+1/2}, \quad S = B^{1/2}, \quad D = LXL, \quad F = LYL \quad (\text{S30})$$

on $P_B\mathbb{C}^n$. Equation (S17) implies

$$Q_0 = (I_3 \otimes S) \begin{pmatrix} D \\ -F \\ P_B \end{pmatrix} (D \quad -F \quad P_B) (I_3 \otimes S). \quad (\text{S31})$$

Choose an orthonormal eigenbasis $De_j = d_j e_j$ on $P_B\mathbb{C}^n$. Since $D = D^*$, all d_j are real. Expanding the middle Gram operator in this basis gives

$$Q_0 = \sum_j |\tilde{\xi}_j\rangle\langle\tilde{\xi}_j|, \quad (\text{S32})$$

$$|\tilde{\xi}_j\rangle = (d_j e_0 + e_2) \otimes Se_j - e_1 \otimes SFe_j. \quad (\text{S33})$$

The Schmidt vectors of $\tilde{\xi}_j$ on $\mathbb{C}^3$ lie in

$$\mathcal{U}_j = \text{span}_{\mathbb{C}}\{d_j e_0 + e_2, e_1\} = (1, 0, -d_j)^{\perp}. \quad (\text{S34})$$

This subspace is invariant under entrywise conjugation and has a real orthonormal basis. Applying the tensor flip gives

$$|\xi_j\rangle := F_{3,n}|\tilde{\xi}_j\rangle = Se_j \otimes (d_j e_0 + e_2) - SFe_j \otimes e_1,$$

which is exactly the form in Eq. (9), with $u_j = Se_j$ and $v_j = SFe_j$. Equations (S29) and (S32) prove positivity and the asserted decomposition of $\tilde{Q}$. Applying the tensor flip termwise gives Eq. (S26) for Q . This includes singular B , and no commutativity of D and F has been used. $\square$

The decomposition is not termwise PPT. The vectors in Eq. (S33) can be entangled, their projectors are then NPT, and Q_0 itself can be NPT. What will matter is that the Schmidt vectors on $\mathbb{C}^3$ of every pure component lie in a two-dimensional subspace invariant under complex conjugation.

S3. PARTIAL-TRANSPOSE INVARIANCE AND DECOMPOSABILITY

Kraus et al. proved that every density operator $Z \in M_2 \otimes M_k$ satisfying $Z^{\Gamma_{\text{qubit}}} = Z$ is separable, where the partial transpose acts on the first, qubit factor [18, Theorem 2]. Normalizing any nonzero positive Z extends the statement to the unnormalized positive cone, while $Z = 0$ is trivial. Their proof works on the minimal support of the second factor and imposes no full-rank assumption.

The same result and duality between the PPT cone and decomposable witnesses imply that every positive map $\Theta : M_2 \rightarrow M_k$ satisfying $\Theta = \Theta \circ T_2$ is decomposable. Indeed, its Choi operator J_Θ is block positive and fixed by partial transpose on its qubit input. For arbitrary $Z \in \text{PPT}(2, k)$, set

$$Z_s = \frac{1}{2}(Z + Z^{\Gamma_{\text{qubit}}}). \quad (\text{S35})$$

Our standing PPT convention transposes the second factor, but $Z^{\Gamma_{\text{qubit}}} = (Z^{\Gamma_{\text{output}}})^T \geq 0$. Thus Z_s is positive, invariant under partial transpose, and separable by the theorem of Kraus et al. [18]. Moreover, PT invariance of J_Θ gives

$$\text{Tr}(J_\Theta Z) = \text{Tr}(J_\Theta Z_s) \geq 0. \quad (\text{S36})$$

Duality between the PPT cone and decomposable witnesses proves the claim.

From this point onward, Q denotes the operator in Eq. (S22). We use its pure-state decomposition to prove the inequality needed in Sec. S4. Define

$$\text{Sym}(Z) = \frac{1}{2}(Z + Z^T). \quad (\text{S37})$$

Lemma S2. *Let Q be the operator from Lemma S1. If $\Lambda : M_3 \rightarrow M_k$ is positive and $\Lambda = \Lambda \circ \text{Sym}$, then for every $P \in \text{PPT}(n, k)$,*

$$\text{Tr}[P(\text{id}_n \otimes \Lambda)(Q)] \geq 0. \quad (\text{S38})$$

Proof. If $U : \mathbb{C}^2 \rightarrow \mathbb{C}^3$ is a real isometry, then

$$\Lambda \circ \text{Ad}_U = (\Lambda \circ \text{Ad}_U) \circ T_2, \quad (\text{S39})$$

because U is real and Λ discards the part antisymmetric under transposition. The cited result for operators invariant under partial transpose and duality between the PPT cone and decomposable witnesses make every such restriction decomposable.

For each term in Eq. (S26), choose a real isometry $U_\alpha : \mathbb{C}^2 \rightarrow \mathbb{C}^3$ whose range contains all Schmidt vectors on $\mathbb{C}^3$, and write

$$\psi_\alpha = (I_n \otimes U_\alpha)\phi_\alpha, \quad \phi_\alpha = \text{vec}(C_\alpha), \quad C_\alpha : \mathbb{C}^n \rightarrow \mathbb{C}^2, \quad (\text{S40})$$

where $\text{vec}(C) = \sum_i e_i \otimes C e_i$. The corresponding term in $(\text{id}_n \otimes \Lambda)(Q)$ is the Choi operator of

$$(\Lambda \circ \text{Ad}_{U_\alpha}) \circ \text{Ad}_{C_\alpha} : M_n \rightarrow M_k. \quad (\text{S41})$$

Decomposable maps remain decomposable under CP precomposition. At the copositive part this follows explicitly from

$$T_2 \circ \text{Ad}_{C_\alpha} = \text{Ad}_{\overline{C_\alpha}} \circ T_n. \quad (\text{S42})$$

Thus every operator obtained from Eq. (S41) pairs nonnegatively with $P \in \text{PPT}(n, k)$. Summing over α proves Eq. (S38). $\square$

S4. POSITIVE MAPS WITH TWO-DIMENSIONAL DOMAIN

For a Hermiticity-preserving map $T : M_2 \rightarrow M_d$, define

$$\mathcal{B}_T(X, R) = \text{Tr}[X(\text{id}_n \otimes T)(R)]. \quad (\text{S43})$$

Lemma S3. *For every positive $T : M_2 \rightarrow M_d$, every $X \in \text{PPT}(n, d)$, and every $R \in \text{PPT}(n, 2)$,*

$$\mathcal{B}_T(X, R) \geq 0. \quad (\text{S44})$$

Proof. The zero map is trivial, so assume $T \neq 0$. The support of $T(I_2)$ contains the range of every output of T : for a positive input Z , $0 \leq Z \leq \|Z\|I_2$, and linearity then covers all inputs. We may therefore restrict the codomain to $\text{supp } T(I_2)$; compressing X to it preserves PPT and does not change the pairing.

Some pure input has nonzero output. An invertible input congruence moves that vector to e_0 . Let $k = \text{rank } T(E_{00}) \geq 1$ and $m = d - k$. The bilinear form is invariant under the congruences

$$T' = \text{Ad}_W \circ T \circ \text{Ad}_U, \quad R' = (\text{id}_n \otimes \text{Ad}_{U^{-1}})(R), \quad X' = (\text{id}_n \otimes \text{Ad}_{W^{-*}})(X), \quad (\text{S45})$$

where U and W are invertible and $W^{-*} = (W^{-1})^*$, then cyclicity gives

$$\mathcal{B}_{T'}(X', R') = \mathcal{B}_T(X, R). \quad (\text{S46})$$

The transformed X' and R' remain PPT because they are obtained by local filters. Thus both input and output congruences may be used without changing the claim. After whitening $T(E_{00})$ on its range, write

$$\begin{aligned} T(E_{00}) &= \begin{pmatrix} I_k & 0 \\ 0 & 0_m \end{pmatrix}, \\ T(E_{01}) &= \begin{pmatrix} H & A \\ B^* & G \end{pmatrix}, \end{aligned} \quad T(E_{11}) = \begin{pmatrix} C & F_0 \\ F_0^* & D_0 \end{pmatrix}. \quad (\text{S47})$$

Within this proof, the letters A, B, C in Eq. (S47) refer only to blocks of T , not to the blocks defining Q in Sec. S2.

For every $z \in \mathbb{C}$, positivity of

$$T(|0 + z1\rangle\langle 0 + z1|) \quad (\text{S48})$$

implies positivity of its lower principal block $\bar{z}G + zG^* + |z|^2 D_0$. Letting the modulus of z tend to zero with arbitrary phase gives $G = 0$, while $T(E_{11}) \geq 0$ gives $D_0 \geq 0$. If $y \in \ker D_0$, positivity of $T(E_{11})$ implies $F_0 y = 0$. The vector $0 \oplus y$ then has zero expectation against every operator in Eq. (S48); positivity forces that operator to annihilate $0 \oplus y$. Its upper component is

$$\bar{z}Ay + zBy. \quad (\text{S49})$$

Taking $z = 1$ and $z = i$ gives $Ay = By = 0$. Hence $0 \oplus y$ is a common kernel vector of all four matrix-unit images, contrary to the assumption $\text{supp } T(I_2) = \mathbb{C}^d$. Therefore

$$D_0 > 0 \quad \text{when } m > 0. \quad (\text{S50})$$

The invertible upper shear with off-diagonal block $-F_0 D_0^{-1}$ removes F_0 without changing the normalized $T(E_{00})$. A lower whitening then gives, after relabeling the remaining blocks,

$$\begin{aligned} T(E_{00}) &= \begin{pmatrix} I_k & 0 \\ 0 & 0_m \end{pmatrix}, & T(E_{01}) &= \begin{pmatrix} H & A \\ B^* & 0 \end{pmatrix}, \\ T(E_{11}) &= \begin{pmatrix} C & 0 \\ 0 & I_m \end{pmatrix}. \end{aligned} \quad (\text{S51})$$

When $m = 0$, all lower blocks, the shear, and the lower whitening are simply absent.

For a real vector $x = (u, v, r)^\top \in \mathbb{R}^3$, define

$$M(u, v) = u(A + B) + iv(B - A), \quad (\text{S52})$$

$$\begin{aligned} \mathcal{S}(u, v, r) &= (u^2 + v^2)I_k - M(u, v)M(u, v)^* \\ &\quad + r[u(H + H^*) + iv(H^* - H)] + r^2 C. \end{aligned} \quad (\text{S53})$$

Since $Vx = (u - iv, r)^\top$, the normal form gives

$$T((Vx)(Vx)^*) = \begin{pmatrix} \mathcal{S}(u, v, r) + M(u, v)M(u, v)^* & rM(u, v) \\ rM(u, v)^* & r^2 I_m \end{pmatrix}. \quad (\text{S54})$$

For $m > 0$ and $r \neq 0$, $\mathcal{S}(u, v, r) \geq 0$ is the Schur complement of the lower-right block. Continuity covers $r = 0$. For $m = 0$, the rectangular matrices A, B, M are empty and $\mathcal{S}(x) = T((Vx)(Vx)^*) \geq 0$ directly.

Polarization gives a unique real-linear map $\Lambda_{\mathbb{R}}$ from real symmetric 3×3 matrices to Hermitian $k \times k$ matrices, characterized by

$$\Lambda_{\mathbb{R}}\left(\frac{xy^\top + yx^\top}{2}\right) = \frac{\mathcal{S}(x + y) - \mathcal{S}(x) - \mathcal{S}(y)}{2}, \quad x, y \in \mathbb{R}^3.$$

Let $\Lambda_{\mathbb{C}}$ be its complexification on complex symmetric matrices and define $\Lambda = \Lambda_{\mathbb{C}} \circ \text{Sym}$. Then $\Lambda : M_3 \rightarrow M_k$ is complex linear and satisfies

$$\Lambda(xx^\top) = \mathcal{S}(x), \quad \Lambda = \Lambda \circ \text{Sym}. \quad (\text{S55})$$

This extension is positive. Indeed, for $Z \geq 0$,

$$\text{Sym}(Z) = \frac{1}{2}(Z + \bar{Z}) \geq 0 \quad (\text{S56})$$

is a real positive matrix and hence a sum of real rank-one matrices $x_\ell x_\ell^\top$; Eq. (S55) maps each summand to a positive operator.

Let $E : \mathbb{C}^k \hookrightarrow \mathbb{C}^d$ be the canonical inclusion into the first k coordinates. For the standard basis e_a of $\mathbb{C}^m$, define

$$K_a = \begin{pmatrix} (A+B)e_a & i(B-A)e_a & 0 \\ 0 & 0 & e_a \end{pmatrix} : \mathbb{C}^3 \rightarrow \mathbb{C}^d, \quad a = 1, \dots, m. \quad (\text{S57})$$

For real $x = (u, v, r)^\top$, one has

$$K_a x = \begin{pmatrix} M(u, v)e_a \\ r e_a \end{pmatrix}. \quad (\text{S58})$$

Summing $(K_a x)(K_a x)^*$ over a and using Eq. (S54) proves the quadratic-form identity; polarization gives the map identity

$$T \circ \text{Ad}_V \circ \text{Sym} = \text{Ad}_E \circ \Lambda + \sum_{a=1}^m \text{Ad}_{K_a} \circ \text{Sym}. \quad (\text{S59})$$

Let Q be the operator constructed from R in Sec. S2. Since $(\text{id}_n \otimes \text{Sym})(Q) = Q$ by Eq. (S23), recovery (S25) and identity (S59) give, with

$$P = (\text{I}_n \otimes E^*)X(\text{I}_n \otimes E) \in \text{PPT}(n, k), \quad (\text{S60})$$

the exact splitting

$$\begin{aligned} \mathcal{B}_T(X, R) &= \text{Tr}[P(\text{id}_n \otimes \Lambda)(Q)] \\ &\quad + \sum_{a=1}^m \text{Tr}[(\text{I}_n \otimes K_a^*)X(\text{I}_n \otimes K_a)Q]. \end{aligned} \quad (\text{S61})$$

The first term is nonnegative by Lemma S2. For each a , both operators inside the second trace are positive, so their trace product is nonnegative. If $m = 0$, the sum is empty and the first term is the whole expression. This proves Eq. (S44) for every rank of $T(E_{00})$, including singular and full-rank cases. $\square$

S5. TWO-DIMENSIONAL INTERMEDIATE ALGEBRA: PAULI REPRESENTATION

Very recently, Henry proved the case with a two-dimensional intermediate algebra using normalized 2×2 Choi blocks and operator interpolation [15]. The proof below instead uses the three Pauli matrices and a separability identity for Hermitian row contractions; it is logically independent of the proof for a two-dimensional input or output algebra.

Lemma S4. *Let $X_1, \dots, X_r \in \text{M}_a$ and $Y_1, \dots, Y_r \in \text{M}_c$ be Hermitian and satisfy*

$$\sum_{j=1}^r X_j^2 \leq \text{I}_a, \quad \sum_{j=1}^r Y_j^2 \leq \text{I}_c. \quad (\text{S62})$$

Then

$$\text{I}_a \otimes \text{I}_c + \sum_{j=1}^r X_j \otimes Y_j \quad (\text{S63})$$

is separable across $\mathbb{C}^a : \mathbb{C}^c$.

Proof. Set $A_j = |X_j|$, $B_j = |Y_j|$, and introduce positive and negative parts

$$P_j = \frac{A_j + X_j}{2}, \quad N_j = \frac{A_j - X_j}{2},$$

$$Q_j = \frac{B_j + Y_j}{2}, \quad R_j = \frac{B_j - Y_j}{2}. \quad (\text{S64})$$

The deficits

$$D_X = I_a - \sum_j A_j^2, \quad D_Y = I_c - \sum_j B_j^2 \quad (\text{S65})$$

are positive. Direct expansion yields

$$I_a \otimes I_c + \sum_j X_j \otimes Y_j = 2 \sum_j (P_j \otimes Q_j + N_j \otimes R_j) + \mathcal{R}, \quad (\text{S66})$$

$$\begin{aligned} \mathcal{R} &= \frac{1}{2} D_X \otimes I_c + \frac{1}{2} I_a \otimes D_Y \\ &\quad + \frac{1}{2} \sum_j (A_j \otimes I_c - I_a \otimes B_j)^2. \end{aligned} \quad (\text{S67})$$

The first two terms of Eq. (S67) are positive products. For each j , choose eigenbases $A_j = \sum_p a_{jp} |p_j\rangle\langle p_j|$ and $B_j = \sum_q b_{jq} |q_j\rangle\langle q_j|$. Then

$$(A_j \otimes I - I \otimes B_j)^2 = \sum_{p,q} (a_{jp} - b_{jq})^2 |p_j\rangle\langle p_j| \otimes |q_j\rangle\langle q_j|, \quad (\text{S68})$$

which is separable. Every term on the right-hand side of Eq. (S66) is therefore separable. $\square$

For the case $d_1 = 2$, let $\Phi_1 : M_{d_0} \rightarrow M_2$ and $\Phi_2 : M_2 \rightarrow M_{d_2}$ be PPT. Put $\Psi = \Phi_1^* : M_2 \rightarrow M_{d_0}$ and $\Theta = \Phi_2 : M_2 \rightarrow M_{d_2}$. First suppose that both maps are unital, and define

$$X_j = \Psi(\sigma_j), \quad Y_j = \Theta(\sigma_j), \quad j = 1, 2, 3, \quad (\text{S69})$$

for the Pauli matrices. Complete positivity of Ψ gives

$$\frac{1}{2} \left(I_2 \otimes I_{d_0} + \sum_{j=1}^3 \sigma_j^\top \otimes X_j \right) \geq 0. \quad (\text{S70})$$

Because Ψ is also completely copositive, the map $\Psi \circ T_2 \circ \text{Ad}_{\sigma_y}$ is CP. The spin-flip identity

$$(T_2 \circ \text{Ad}_{\sigma_y})(\sigma_j) = -\sigma_j, \quad (T_2 \circ \text{Ad}_{\sigma_y})(I_2) = I_2 \quad (\text{S71})$$

therefore gives Eq. (S70) with the minus sign. Hence, for $H_X = \sum_j \sigma_j^\top \otimes X_j$, $-I \leq H_X \leq I$, and thus $H_X^2 \leq I$. Taking the normalized qubit partial trace and using $\text{Tr}(\sigma_j \sigma_k) = 2\delta_{jk}$ gives

$$\sum_{j=1}^3 X_j^2 = \frac{1}{2} \text{Tr}_{\mathbb{C}^2}(H_X^2) \leq I_{d_0}. \quad (\text{S72})$$

The same argument for Θ gives $\sum_j Y_j^2 \leq I_{d_2}$. Expansion in the orthogonal Pauli basis now yields

$$J_{\Theta \circ \Psi^*} = \frac{1}{2} \left(I_{d_0} \otimes I_{d_2} + \sum_{j=1}^3 X_j^\top \otimes Y_j \right), \quad (\text{S73})$$

which is separable by Lemma S4.

It remains to remove unitality without assuming full rank. If either map is zero there is nothing to prove. Otherwise let

$$S_0 = \Psi(I_2), \quad P_0 = \text{supp } S_0, \quad S_2 = \Theta(I_2), \quad P_2 = \text{supp } S_2, \quad (\text{S74})$$

and regard $P_i \mathbb{C}^{d_i}$ as the corresponding output space. Define

$$\widehat{\Psi}(Z) = S_0^{+1/2} \Psi(Z) S_0^{+1/2}, \quad B_0 = S_0^{1/2} : P_0 \mathbb{C}^{d_0} \rightarrow \mathbb{C}^{d_0},$$

$$\widehat{\Theta}(Z) = S_2^{+/2} \Theta(Z) S_2^{+/2}, \quad B_2 = S_2^{1/2} : P_2 \mathbb{C}^{d_2} \rightarrow \mathbb{C}^{d_2}. \quad (\text{S75})$$

Positivity implies that every output of Ψ , respectively Θ , is supported on P_0 , respectively P_2 . Thus the hatted maps are unital on their support spaces and

$$\Psi = \text{Ad}_{B_0} \circ \widehat{\Psi}, \quad \Theta = \text{Ad}_{B_2} \circ \widehat{\Theta}. \quad (\text{S76})$$

They remain PPT: output transposition changes the whitening filter $S_i^{+/2}$ into $\overline{S_i^{+/2}}$. Taking adjoints in the first identity gives

$$\Theta \circ \Psi^* = \text{Ad}_{B_2} \circ (\widehat{\Theta} \circ \widehat{\Psi}^*) \circ \text{Ad}_{B_0^*}. \quad (\text{S77})$$

The unital case makes the Choi operator of $\widehat{\Theta} \circ \widehat{\Psi}^*$ separable. The two outside CP congruences act locally on that Choi operator and preserve separability, completing the alternative proof.

S6. CHOI COMPOSITION AND CONSEQUENCES

We first use Lemma S3 to prove the composition statement with a two-dimensional output algebra. Let $\Phi : M_d \rightarrow M_n$ and $\Psi : M_n \rightarrow M_2$ be PPT. Then

$$P = J_{\Phi^*} \in \text{PPT}(n, d), \quad R = J_{\Psi} \in \text{PPT}(n, 2), \quad (\text{S78})$$

because adjoints preserve complete positivity and complete copositivity. For every positive $T : M_2 \rightarrow M_d$, a matrix-unit expansion gives

$$\begin{aligned} \mathcal{B}_T(J_{\Phi^*}, J_{\Psi}) &= \text{Tr}_{\text{HS}}(T \circ \Psi \circ \Phi) \\ &= \text{Tr}[J_{T^*} J_{\Psi \circ \Phi}], \end{aligned} \quad (\text{S79})$$

where, for an endomorphism $S : M_d \rightarrow M_d$,

$$\text{Tr}_{\text{HS}} S = \sum_{i,j} \text{Tr}[E_{ji} S(E_{ij})]. \quad (\text{S80})$$

The first equality also uses cyclicity of the finite-dimensional superoperator trace. As T ranges over all positive maps $M_2 \rightarrow M_d$, J_{T^*} ranges over the cone of all block-positive operators on $\mathbb{C}^d \otimes \mathbb{C}^2$. Equations (S44) and (S79), together with block-positive/separable duality, therefore imply

$$J_{\Psi \circ \Phi} \in \text{SEP}(d; 2). \quad (\text{S81})$$

Hence

$$\text{PPT}(M_n, M_2) \circ \text{PPT}(M_d, M_n) \subseteq \text{EB}(M_d, M_2). \quad (\text{S82})$$

Ordinary EB maps are closed under Hilbert–Schmidt adjoints: the Choi operator of the adjoint is obtained from the original one by factor swap and entrywise conjugation, which preserve separability. Taking adjoints therefore gives the case with a two-dimensional input algebra,

$$\text{PPT}(M_n, M_d) \circ \text{PPT}(M_2, M_n) \subseteq \text{EB}(M_2, M_d). \quad (\text{S83})$$

Here the order is inner factor first; the intermediate dimension n and the other outer dimension d are independent.

Equation (S82) proves the case in which the output algebra is two dimensional, while Eq. (S83) gives the case of a two-dimensional input algebra by adjunction. Combining them with the proof for a two-dimensional intermediate algebra in Sec. S5 proves Theorem 1 for $d_0 = 2$, $d_1 = 2$, or $d_2 = 2$. Explicitly, if

$$\Phi_1 : M_{d_0} \rightarrow M_{d_1}, \quad \Phi_2 : M_{d_1} \rightarrow M_{d_2} \quad (\text{S84})$$

are PPT and $2 \in \{d_0, d_1, d_2\}$, then

$$\Phi_2 \circ \Phi_1 \in \text{EB}(M_{d_0}, M_{d_2}). \quad (\text{S85})$$

The other dimensions need not coincide.

Equation (S9) gives the stronger consequence for states without requiring the two relay dimensions to coincide. Namely, if $\rho_{AB} \in \text{PPT}(a, b)$ and $\sigma_{CD} \in \text{PPT}(c, d)$, and if $2 \in \{a, b, c, d\}$, then for every positive operator $M_{BC} \geq 0$,

$$\omega_M(\rho, \sigma) \in \text{SEP}(A:D). \quad (\text{S86})$$

After either identity in Eq. (S9) is applied, the two filtered states are Choi operators of composable PPT maps, and one of the three matrix algebras has dimension two. Equation (S10) preserves PPT under the required filters, and spectral decomposition covers arbitrary $M \geq 0$.

Following Ref. [5, Definition II.2], a CP map $H : M_a \rightarrow M_b$ is 2-entanglement breaking (2-EB) when $(\text{id}_2 \otimes H)(Z)$ is separable for every $Z \geq 0$ in $M_2 \otimes M_a$. Equivalently, $H \circ E$ is EB for every CP map $E : M_2 \rightarrow M_a$ [20, Theorem 3.5].

Proof of Corollary 6.— For arbitrary composable PPT maps $\Phi_1 : M_{d_0} \rightarrow M_{d_1}$ and $\Phi_2 : M_{d_1} \rightarrow M_{d_2}$, put $H = \Phi_2 \circ \Phi_1$. For arbitrary CP maps $E_i : M_2 \rightarrow M_{d_i}$, $i = 0, 2$, the maps $\Phi_1 \circ E_0$ and $\Phi_2^* \circ E_2$ are PPT, because PPT is preserved under adjoints and CP precomposition. The case $d_0 = 2$ of Theorem 1 makes $H \circ E_0$ and $H^* \circ E_2$ EB. The equivalent characterization above proves both 2-EB assertions separately.

Chains and iteration.— Any composition of at least two composable PPT maps is EB if one space along the chain is a qubit. For $d \geq 2$, every composition of $2(d-1)$ PPT maps on a fixed M_d is EB; in particular,

$$T^{2(d-1)} \in \text{EB}(M_d, M_d) \quad (\text{S87})$$

for every PPT map $T : M_d \rightarrow M_d$. PPT maps are closed under composition. If the qubit is internal to a chain, compose all maps on either side into two PPT maps and apply the case $d_1 = 2$ of Theorem 1. If it is an input or output algebra, retain the adjacent factor and compose the remaining nonempty side, then apply the corresponding case of the theorem.

For maps on a fixed M_d , group the $2(d-1)$ factors into $d-1$ adjacent pairs. Each $T_{2j} \circ T_{2j-1}$ is 2-EB by Corollary 6. The Schmidt-number iteration theorem of Christandl, Müller-Hermes, and Wolf states that a composition of $d-1$ 2-EB maps on M_d is entanglement breaking [5, Theorem II.1], proving the claim. Section VI of Ref. [5] proved the following conditional implication: if T^2 is 2-EB for every PPT map T , then $T^{2(d-1)}$ is EB. Corollary 6 verifies the required 2-EB property and proves more: it applies to every composable pair of PPT maps, so the adjacent factors need not be identical.

Writing $n_{\text{EB}}(T) = \min\{n \geq 1 : T^n \in \text{EB}\}$, Eq. (S87) gives the uniform bound

$$\sup_{T \in \text{PPT}(M_d, M_d)} n_{\text{EB}}(T) \leq 2(d-1), \quad (\text{S88})$$

and therefore answers Question 4.3 of Ref. [12]. The statement for a chain of different PPT maps is stronger than the bound on powers of a single map.

The iteration result does not settle the PPT squared conjecture for arbitrary pairs in equal dimensions $d \geq 4$: a general pair is shown here to have a 2-EB composition, not necessarily an entanglement-breaking one.